Introductory Statistics for Econometrics

Unit 5. Covariance, Conditional Expectation and Random Vectors

Covariance, correlation, conditional expectation, the law of iterated expectations, mean vectors and covariance matrices

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Learning outcomes. After this unit the student should be able to (i) compute and interpret covariance and the correlation coefficient; (ii) state the variance of a linear combination of (possibly dependent) RVs; (iii) work with the mean vector and covariance matrix of a random vector; (iv) compute conditional expectations and conditional variances; (v) apply the law of iterated expectations and the law of total variance; (vi) recognise the OLS coefficient as the linear projection of Y on X.

Source material: condensed from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 13–14, augmented with the multivariate framework.

1 Covariance

The variance of a single random variable measured its dispersion. The *covariance* measures the joint variation of a pair.

Definition 1.1 (Covariance)**.** For random variables X, Y with finite second moments,

$$\mathrm{Cov}(X,Y) \;=\; \mathbb{E}\big[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])\big].$$

Proposition 1.2 (Algebra of covariance)**.** *(1)* $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$*.*

(2) Symmetry: $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$*.*

(3) Computational formula: $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$*.*

(4) Bilinearity: $\mathrm{Cov}(X,\, aY + bZ + c) = a\mathrm{Cov}(X,Y) + b\mathrm{Cov}(X,Z)$*.*

(5) Affine invariance: $\mathrm{Cov}(aX+b,\, cY+d) = ac\,\mathrm{Cov}(X,Y)$*.*

(6) If $X \perp Y$ *then* $\mathrm{Cov}(X,Y) = 0$*. The converse is* false *in general.*

Proposition 1.3 (Variance of a sum)**.** *For any X, Y with finite second moments,*

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y).$$

More generally, for any constants a_i,

$$\mathrm{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \mathrm{Var}(X_i) + 2\sum_{i<j} a_i a_j \mathrm{Cov}(X_i, X_j).$$

Example 1.4. Let $f_{XY}(x, y) = 8xy$ on $\{0 \leq x \leq y \leq 1\}$, zero elsewhere. Direct LOTUS calculations give $\mathbb{E}[X] = 8/15$, $\mathbb{E}[Y] = 4/5$ and $\mathbb{E}[XY] = 4/9$, so

$$\mathrm{Cov}(X, Y) \;=\; \frac{4}{9} - \frac{8}{15} \cdot \frac{4}{5} \;=\; \frac{4}{225} > 0.$$

The positive covariance is consistent with the support: Y is constrained above X, so larger X goes with larger Y.

2 Correlation coefficient

Covariance has the units of $X \cdot Y$, so it changes when we rescale. The *correlation coefficient* normalises away the scale.

Definition 2.1 (Correlation coefficient)**.** For $\mathrm{Var}(X), \mathrm{Var}(Y) > 0$,

$$\rho(X, Y) \;\equiv\; \mathrm{Corr}(X, Y) \;=\; \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}.$$

Proposition 2.2. *(i)* $-1 \leq \rho(X, Y) \leq 1$.

(ii) $|\rho(X, Y)| = 1$ *iff* $Y = aX + b$ *for some constants* $a \neq 0, b$.

(iii) $\rho(aX + b,\, cY + d) = \mathrm{sign}(ac) \cdot \rho(X, Y)$.

Remark 2.3 (Correlation $\neq$ causation). Covariance and correlation are symmetric in X and Y — they cannot tell "what causes what". Time spent at the gym and self-reported health are positively correlated; this does not establish that exercise causes better health (reverse causation, omitted variables, selection — your econometrics course is largely about distinguishing these). Causation requires extra structure (an experiment, an instrument, a natural discontinuity), not just a positive correlation.

Remark 2.4 (Zero correlation $\not\Rightarrow$ independence). Let $X \sim U[-1, 1]$ and $Y = X^2$. Then $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0$ by symmetry, so $\mathrm{Cov}(X, Y) = 0$. But Y is a deterministic function of X, so X and Y are far from independent. The correlation captures only the *linear* part of dependence.

3 Mathematical introduction to vector and matrix notation

Definition 3.1. A *matrix* is a rectangular array or grid of numbers, symbols, or expressions arranged in rows and columns. It serves as a foundational tool in mathematics for data organization, solving linear equations, and performing transformations.

In general we denote the dimensions of the matrix (number of rows and number of columns) as a subindex of the matrix. As a convention we first write the number of rows and then the number of columns, $A_{m \times n}$.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

Equivalently, you will also see this $A \in \mathbb{R}^{m \times n}$. This notation gives you some extra information, first it tells you the number of rows is m and the number of columns is n. Secondly, it tells you that each of the elements of the matrix a_{ij} are real numbers.

A matrix is called *square* if it has the same number of rows and columns. That is,

$$A \in \mathbb{R}^{n \times n}.$$

We will say a matrix is diagonal if it is square and all of the elements except of the ones in the main diagonal are zero. Using as example the previous matrix we will say A is diagonal if

$$\begin{cases} a_{ii} \neq 0 & \text{for some i} \\ a_{ij} = 0 & \text{for all } i \neq j \end{cases}$$

Definition 3.2. A *vector* is just a matrix that has one of its dimensions equal to 1. Therefore we have two definitions of vectors

Column vector: The number of columns is equal to 1

$$A_{3\times 1} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

Row vector: The number of rows is equal to 1

$$A_{1\times 3} = \begin{pmatrix} a_1, & a_2, & a_3 \end{pmatrix}$$

Row vectors include commas in order to ensure the different entrances. As we can see, whether we represent a vector as a column vector or as a row vector is mostly a matter of convention. Both representations can be used for the same purpose, provided that the definitions are used consistently.

In many cases, when introducing a vector, we do not explicitly specify whether it is a row or a column vector. Sometimes we only state its dimension, for example

$$A \in \mathbb{R}^3.$$

Throughout this course, unless stated otherwise, vectors will always be treated as **column vectors**. From now on vectors will be presented in bold letters.

Definition 3.3. The *transpose* represents a change in the relative order, position, or sequence of items, often by swapping them. The symbol is $\top$ or just $'$ at the right top of the matrix or the vector. Consider a matrix $A_{m\times n}$

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

and a vector B of dimension k

$$B = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_k \end{pmatrix}$$

then,

$$A^\top = A' = \begin{pmatrix} a_{11} & a_{21} & \dots & a_{n1} \\ a_{12} & a_{22} & \dots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{pmatrix}; \quad B^\top = B' = (b_1, b_2, \dots, b_k)$$

Notice that $A^\top$ is a matrix of dimension $n \times m$ and $B^\top$ is a row vector (dimension $1 \times k$.

We will use this to simplify notation but you must use it correctly **remember from your previous studies** that there are some rules regarding multiplication of matrices. You should check this before the start of the econometrics course because it will be useful in order to have a good use of matrix notation.

The following properties will be useful when working with matrix notation.

- **Matrix equality.** Two matrices are equal if they have the same dimension and all their corresponding entries are equal. That is, if

  $$A, B \in \mathbb{R}^{m\times n},$$

  then

  $$A = B \quad \text{implies that} \quad a_{ij} = b_{ij} \quad \text{for all } i = 1, \dots, m,\ j = 1, \dots, n.$$

- **Matrix addition.** Matrix addition is only defined for matrices with the same dimension. If

  $$A, B \in \mathbb{R}^{m\times n},$$

  then

  $$A + B \in \mathbb{R}^{m\times n},$$

  and each element is given by

  $$(A + B)_{ij} = a_{ij} + b_{ij}.$$

- **Scalar multiplication.** If $A \in \mathbb{R}^{m\times n}$ and $\lambda \in \mathbb{R}$, then

  $$\lambda A \in \mathbb{R}^{m\times n},$$

  where

  $$(\lambda A)_{ij} = \lambda a_{ij}.$$

- **Matrix multiplication.** Matrix multiplication is only defined when the number of columns of the first matrix is equal to the number of rows of the second matrix. If

  $$A \in \mathbb{R}^{m\times n} \quad \text{and} \quad B \in \mathbb{R}^{n\times k},$$

  then

  $$AB \in \mathbb{R}^{m\times k}.$$

  The (i, j)-th element of AB is

  $$(AB)_{ij} = \sum_{\ell=1}^{n} a_{i\ell} b_{\ell j}.$$

- **Matrix multiplication is not generally commutative.** In general,

  $$AB \neq BA.$$

  In fact, AB and BA may not even both be well defined. Therefore, the order of multiplication matters.

- **Associativity of matrix multiplication.** If the products are well defined, then

  $$A(BC) = (AB)C.$$

- **Distributive property.** If the dimensions are compatible, then

$$A(B + C) = AB + AC,$$

and

$$(A + B)C = AC + BC.$$

- **Identity matrix.** The identity matrix of dimension $n \times n$ is denoted by I_n. It has ones on the main diagonal and zeros elsewhere:

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}.$$

If $A \in \mathbb{R}^{m \times n}$, then

$$I_m A = A \quad \text{and} \quad AI_n = A.$$

- **Zero matrix.** The zero matrix is a matrix whose entries are all equal to zero. If $A \in \mathbb{R}^{m \times n}$, then

$$A + 0_{m \times n} = A.$$

- **Transpose of a sum.** If $A, B \in \mathbb{R}^{m \times n}$, then

$$(A + B)^\top = A^\top + B^\top.$$

- **Transpose of a scalar multiple.** If $A \in \mathbb{R}^{m \times n}$ and $\lambda \in \mathbb{R}$, then

$$(\lambda A)^\top = \lambda A^\top.$$

- **Transpose of a product.** If the product AB is well defined, then

$$(AB)^\top = B^\top A^\top.$$

Notice that the order of multiplication is reversed.

- **Double transpose.** Taking the transpose twice gives back the original matrix:

$$(A^\top)^\top = A.$$

- **Symmetric matrix.** A square matrix $A \in \mathbb{R}^{n \times n}$ is symmetric if

$$A^\top = A.$$

- **Inverse matrix.** A square matrix $A \in \mathbb{R}^{n \times n}$ is invertible if there exists a matrix $A^{-1} \in \mathbb{R}^{n \times n}$ such that

$$A^{-1}A = AA^{-1} = I_n.$$

Not every square matrix has an inverse.

- **Inverse of a product.** If A and B are invertible square matrices of compatible dimensions, then

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Again, the order is reversed.

- **Transpose of an inverse.** If A is invertible, then

$$(A^{-1})^\top = (A^\top)^{-1}.$$

4 Random vectors: mean vector and covariance matrix

When we move from two random variables to many — wages, schooling, experience, region, ... — it is convenient to stack them in a vector.

Definition 4.1 (Random vector). An $n \times 1$ *random vector* is $\boldsymbol{X} = (X_1, \ldots, X_n)^\top$.

Definition 4.2 (Mean vector and covariance matrix).

$$\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{X}] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_n] \end{pmatrix}, \qquad \Sigma = \operatorname{Var}(\boldsymbol{X}) = \mathbb{E}\big[(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^\top\big].$$

The (i, j)-entry of Σ is $\operatorname{Cov}(X_i, X_j)$. The diagonal of Σ holds the marginal variances; off-diagonal entries the covariances.

Proposition 4.3 (Algebra of mean and covariance). *For any constants $A \in \mathbb{R}^{m\times n}$, $\boldsymbol{b} \in \mathbb{R}^m$, and any random vector $\boldsymbol{X}$ with mean $\boldsymbol{\mu}$ and covariance Σ:*

(1) $\mathbb{E}[A\boldsymbol{X} + \boldsymbol{b}] = A\boldsymbol{\mu} + \boldsymbol{b}$.

(2) $\operatorname{Var}(A\boldsymbol{X} + \boldsymbol{b}) = A\,\Sigma\,A^\top$.

(3) Σ is symmetric and positive semi-definite (i.e. $\boldsymbol{a}^\top\Sigma\boldsymbol{a} \geq 0$ for every $\boldsymbol{a} \in \mathbb{R}^n$, with equality iff $\boldsymbol{a}^\top\boldsymbol{X}$ is a constant a.s.).

(4) If $X_1, \ldots, X_n$ are mutually independent, Σ is diagonal.

Example 4.4 (Two-asset portfolio). Let R_1, R_2 be the returns on two assets with $\operatorname{Var}(R_i) = \sigma_i^2$ and $\operatorname{Corr}(R_1, R_2) = \rho$. A portfolio with weights $\boldsymbol{w} = (w_1, w_2)^\top$ has variance

$$\operatorname{Var}(\boldsymbol{w}^\top\boldsymbol{R}) = \boldsymbol{w}^\top\Sigma\boldsymbol{w} = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\,\rho\,\sigma_1\sigma_2.$$

For $w_1 + w_2 = 1$, the variance-minimising weight is $w_1^* = (\sigma_2^2 - \rho\sigma_1\sigma_2)/(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)$, the textbook result on diversification.

Remark 4.5 (A bridge to econometrics). With matrices in hand, OLS is one line: $\widehat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top\boldsymbol{Y}$. The matrices $X^\top X$ and $X^\top Y$ are sample analogues of the covariance algebra in Proposition 4.3. We will not unpack OLS in this unit, but every property here will reappear there.

5 Conditional expectation

When we know that $X = x$, the relevant distribution for Y is the conditional distribution $f_{Y|X}(\cdot \mid x)$ (Unit 3). Its mean is the *conditional expectation*.

Definition 5.1 (Conditional expectation). For $f_X(x) > 0$,

$$\mathbb{E}[Y \mid X = x] = \begin{cases} \sum_y y\, f_{Y|X}(y \mid x), & Y \text{ discrete}, \\ \int_{-\infty}^{\infty} y\, f_{Y|X}(y \mid x)\, dy, & Y \text{ continuous}. \end{cases}$$

The function $x \mapsto \mathbb{E}[Y \mid X = x]$ is the *conditional expectation function* (CEF). When the conditioning variable is left random, $\mathbb{E}[Y \mid X]$ is itself a random variable: a function of X.

The CEF is the structural object of regression analysis: it summarises "how does the average of Y change with x?"

Example 5.2 (Akerlof's market for "lemons"). Three equally likely qualities (lemon, average, melon) of a used car; buyer and seller have valuations

	Seller	Buyer
Lemon	5,000	6,000
Average	6,000	10,000
Melon	10,000	11,000

Without information, $\mathbb{E}[Y_S] = 7000 < 9000 = \mathbb{E}[Y_B]$, so trade happens. But when sellers know the type and post a price, the buyer should condition on "a seller willing to sell at this price". If the price is below \$6,000 the buyer infers "lemon", so $\mathbb{E}[Y_B \mid Y_S < 6000] = 6000$. For "melons", sellers ask $\geq$ \$10000 but $\mathbb{E}[Y_B \mid Y_S \leq 10000] = 9000 < 10000$, so high-quality cars do not trade. The market unravels — *adverse selection*, and a famous example of how conditional expectations capture economic information frictions.

Example 5.3 (Compound binomial). A firm produces X innovations per year with $\mathbb{E}[X] = 2$, $\text{Var}(X) = 2$. Each is a commercial success with probability $p = 0.2$ independently. Let S be the number of successes. Conditional on $X = x$, $S \mid X = x \sim B(x, p)$, so $\mathbb{E}[S \mid X = x] = xp$. Hence $\mathbb{E}[S \mid X] = Xp$, a function of X.

6 Law of iterated expectations

The CEF is connected to the unconditional mean by the most-used identity in econometrics.

Theorem 6.1 (Law of iterated expectations, LIE). *For any random variables X, Y with $\mathbb{E}|Y| < \infty$,*

$$\mathbb{E}[Y] \;=\; \mathbb{E}\big[\mathbb{E}[Y \mid X]\big].$$

Sketch. Let $g(x) = \mathbb{E}[Y \mid X = x]$. By LOTUS,

$$\mathbb{E}[g(X)] = \int g(x) f_X(x)\,dx = \iint y\,\frac{f_{XY}(x,y)}{f_X(x)} f_X(x)\,dy\,dx = \iint y\,f_{XY}(x,y)\,dy\,dx = \mathbb{E}[Y]. \quad \square$$

Example 6.2 (Compound binomial, continued). Using the LIE, $\mathbb{E}[S] = \mathbb{E}[\mathbb{E}[S \mid X]] = \mathbb{E}[Xp] = p\,\mathbb{E}[X] = 0.2 \cdot 2 = 0.4$.

Remark 6.3 (A more general identity). For any function g of X alone, $\mathbb{E}[g(X)\,Y] = \mathbb{E}[\,g(X)\,\mathbb{E}[Y \mid X]\,]$. This is the *taking out what's known* property and underlies most manipulations of conditional expectations in econometrics.

7 Conditional variance and the law of total variance

Definition 7.1 (Conditional variance). $\text{Var}(Y \mid X) = \mathbb{E}\big[(Y - \mathbb{E}[Y \mid X])^2 \mid X\big] = \mathbb{E}[Y^2 \mid X] - (\mathbb{E}[Y \mid X])^2$.

Theorem 7.2 (Law of total variance).

$$\text{Var}(Y) \;=\; \mathbb{E}\big[\text{Var}(Y \mid X)\big] + \text{Var}\big(\mathbb{E}[Y \mid X]\big).$$

The first term is the *unexplained* variation of Y left within each value of X; the second is the *explained* variation of Y across values of X. This decomposition is the population analogue of the ANOVA / R^2 decomposition in regression.

8 The OLS coefficient as a linear projection

We close the unit with a brief preview of how the tools above produce the ordinary-least-squares coefficient. The full discussion is Unit 9.

Suppose we want to approximate Y by an affine function of X, $\widehat{Y} = \alpha + \beta X$, by minimising the mean squared error $L(\alpha, \beta) = \mathbb{E}[(Y - \alpha - \beta X)^2]$. Differentiating and using linearity of expectation,

$$\partial_\alpha L = -2\mathbb{E}[Y - \alpha - \beta X] = 0, \qquad \partial_\beta L = -2\mathbb{E}[X(Y - \alpha - \beta X)] = 0.$$

The first equation gives $\alpha = \mathbb{E}[Y] - \beta\mathbb{E}[X]$. Substituting into the second,

$$\boxed{\beta \;=\; \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}, \qquad \alpha \;=\; \mathbb{E}[Y] - \beta\,\mathbb{E}[X].} \tag{1}$$

The error $U = Y - \alpha - \beta X$ satisfies $\mathbb{E}[U] = 0$ and $\mathrm{Cov}(X, U) = 0$ *by construction.*

Remark 8.1 (Read this twice). Equation (1) is the population OLS coefficient. It is a ratio of the same two objects we have been computing throughout this unit: a covariance and a variance. The empirical OLS estimator simply replaces Cov and Var by their sample analogues. We will return to this in Unit 9.

Summary

- *Covariance*: $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$. Bilinear in its arguments. $X \perp Y \Rightarrow \mathrm{Cov} = 0$ but not conversely.
- *Correlation*: $\rho = \mathrm{Cov}/\sqrt{\mathrm{Var}\mathrm{Var}} \in [-1, 1]$; $|\rho| = 1$ iff Y is an affine function of X.
- *Random vectors*: mean vector $\boldsymbol{\mu}$, covariance matrix Σ symmetric and positive semi-definite. For affine $A\boldsymbol{X} + \boldsymbol{b}$: mean $A\boldsymbol{\mu} + \boldsymbol{b}$, covariance $A\Sigma A^\top$.
- *Conditional expectation*: $\mathbb{E}[Y \mid X = x] = \int y\, f_{Y|X}(y \mid x)\, dy$. The CEF $x \mapsto \mathbb{E}[Y \mid X = x]$ is the structural object of regression.
- *Law of iterated expectations*: $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]]$.
- *Law of total variance*: $\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X])$.
- *Population OLS*: $\beta = \mathrm{Cov}(X,Y)/\mathrm{Var}(X)$ and $\alpha = \mathbb{E}[Y] - \beta\mathbb{E}[X]$ are the coefficients of the linear projection of Y on X.

Suggested exercises

1. Let $X \sim U[-1, 1]$ and $Y = X^2$. Show that $\mathrm{Cov}(X, Y) = 0$ but X and Y are not independent. Compute $\mathbb{E}[Y \mid X]$ and verify the law of iterated expectations.
2. If c is a constant and X is a random variable use the properties of expectation to show that $\mathrm{Cov}(c, X) = 0$, this is the covariance of a random variable with a constant is equal to zero. Give an intuition using the definition of covariance of why this happens.
3. Two assets have returns R_1, R_2 with $\sigma_1 = 0.20$, $\sigma_2 = 0.15$ and $\rho = 0.3$. Find the variance-minimising portfolio weights subject to $w_1 + w_2 = 1$. What if $\rho = -0.3$?
4. For $f_{XY}(x, y) = 8xy$ on $0 \le x \le y \le 1$ (Example 1.4), compute the conditional expectation function $\mathbb{E}[Y \mid X = x]$ and verify the LIE.

5. (Compound expectation.) Let $X \sim B(n, p)$ and conditional on $X = x$, $Y \mid X = x \sim B(x, q)$. Find $\mathbb{E}[Y]$ and $\text{Var}(Y)$ using the LIE and the law of total variance.

6. Show that $\text{Var}(Y \mid X)$ is in general not equal to $\text{Var}(Y) - \text{Var}(\mathbb{E}[Y \mid X])$ point-by-point — only the average is equal. Give a numerical example.

7. (Population OLS.) Let $X \sim U[0, 1]$ and $Y = 1 + 2X + V$ where $V \sim N(0, \sigma^2)$ is independent of X. Compute $\text{Cov}(X, Y)/\text{Var}(X)$ and confirm $\beta = 2$, $\alpha = 1$.